

Article

Lane Optimization of Highway Reconstruction and Expansion Work Zone Considering Carbon Dioxide Emission Factors

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Abstract: During the highway reconstruction and expansion, some lanes are often closed on the construction side to ensure that the construction is carried out normally. The presence of the work zone increases the traffic pressure on the construction side of the highway, causing traffic congestion, increased CO₂ emissions from motor vehicles, and increasing environmental pollution. The bi-level programming model was developed based on the objective of minimizing the travel time and total CO₂ emissions of the system so as to solve it using a quantum particle swarm algorithm with high convergence speed and high intelligence to form the lane optimization scheme for the three forms of reclosing and expanding six-lane highways in both directions. The results show that reasonable use of opposite non-construction lanes in the work zone of a partially closed highway expansion can reduce the total system travel cost, alleviate traffic congestion, reduce CO₂ emissions, and contribute to the sustainable development of transportation, as well as the environment.

Keywords: sustainable development of transportation and environment; the work zone of highways; the lane optimization scheme; the bi-level programming model; the quantum particle swarm algorithm; CO₂ emissions



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1. Introduction

Green and low-carbon is essential to achieving global sustainable development. As an important source of carbon emissions, the transportation industry needs to study further on new ideas and strategies for energy conservation and emission reduction [1]. Currently, CO₂ emissions from the transportation industry mainly come from motor vehicle exhaust, and the congestion level of traffic is a critical factor, affecting the fuel consumption and CO₂ emission level of vehicles [2].

Vehicles on highways travel faster and often have lower vehicle fuel consumption levels than vehicles traveling on urban roads. However, the highway reconstruction and expansion work zones, as a venue for highway construction, reconstruction, maintenance and upgrading, often closes some lanes to ensure regular construction, which leads to longer travel time for vehicles, causing traffic congestion [3], which, in turn, increases emissions. The road traffic conflicts and congestion in the operation zone are mainly due to the reduction in the number of lanes, and the vehicles are forced to merge forcibly and block lanes while increasing the potential danger of accidents [4,5]. This negative effect will be exacerbated by the increase in the length, duration, and volume of traffic in the work zone [6]. In addition, the higher saturation of traffic flow in the work zone will significantly increase vehicle fuel consumption and CO₂ emission levels, bringing about greater energy consumption and pollution [7].

If we want to ensure the standard and rapid passage of social vehicles in the work zone during the highway reconstruction and expansion, the relevant management departments widely adopt the traffic organization form of borrowing the opposite lane to drive [4]. Due to the particularity of the reconstruction and expansion of the highway, the traffic capacity

on one side of the road is significantly reduced. Therefore, when the opposite traffic flow is in a free flow state, a reversible lane is set up. Traffic congestion in the direction of traffic flow in the construction zone will significantly improve road capacity, reduce travel time costs, reduce traffic congestion, and reduce traffic pollution emissions in the work zone [8]. Therefore, in the closed section of the highway construction, reversible lanes with the help of some opposite lanes are an effective way to solve the highway traffic congestion during the highway construction and the CO₂ emissions generated by the congestion.

In this paper, we construct and solve a bi-level programming model with the optimization objectives of minimizing the total system travel cost and minimizing the total system CO₂ emission. Based on the classical Sioux Falls network in the field of traffic assignment, simulation experiments are carried out on three typical lane closure forms in a two-way six-lane highway work zone, and lane optimization schemes are given for three closure conditions to achieve the objectives of the lowest total system travel cost and lowest total system CO₂ emission cost. It provides new ideas and technical means for the subsequent sustainable research of the traffic and the environment in highway work zones.

2. Literature Review

The Highway Capacity Manual 1985 divided the layout form of highway reconstruction work zones by the number of lane closures for reconstruction work zones, identifying six types of reconstruction work zones with two lane closures of 1 lane, three lane closures of one lane, three lane closures of two lanes, four lane closures of two lanes, four lane closures of three lanes, and five lane closures of two lanes. Investigations were conducted for the upstream lane closures of the highway work zone and the form maximum flows of the six layout forms of the reconfiguration work zone of 3-1, 2-1, 5-2, 4-2, 3-2, and 4-3 were determined to be 1170 pcu/h/ln, 1340 pcu/h/ln, 1370 pcu/h/ln, 1480 pcu/h/ln, 1490 pcu/h/ln, and 1520 pcu/h/ln; the closure of some lanes in the work zone will cause traffic congestion, compared to the typical roadway, leading to the occurrence of traffic congestion, increasing the travel time and reducing the efficiency of vehicle driving in the zone [9]. The Highway Capacity Manual 2000 further states that the impact of construction intensity on relieving the maximum traffic volume is $\pm 10\%$ [10]. Rouphail et al. [11], found that the change in vehicle driving speed is not significant at low traffic volumes. However, with increasing traffic volumes, the time required for vehicle driving speed recovery will become significantly longer. Kim et al. [12] proposed that in certain construction sections under road conditions, the factors that affect the capacity of the work zone section are mainly traffic volume, traffic composition, closed lane width and the number of lanes, road longitudinal slope, and driver driving level, with the increase in traffic volume and the number of lanes closed for construction greatly affecting the capacity. Serio [13] enriched the case studies on road engineering under partial lane closure conditions with microsimulations to evaluate the possible impact of implementing road engineering warnings and lane closures on Italian freeways and explored the impact of an innovative system, such as C-ITS, on traffic flow under lane closure conditions.

In addition to the restricted travel efficiency, CO₂ emissions from vehicles in the work zone also increase significantly with frequent acceleration and deceleration. Samaras [14] obtained CO₂ emissions from motor vehicles at different traffic flow densities through field tests and re-corrected the CO₂ emission simulation software based on the test data. The prediction results from the corrected model showed that motor vehicles at higher traffic flow densities presented congested flow. De Vlieger et al. [15,16] found that emissions under actual traffic conditions differed significantly from driver to driver, with forced merging and aggressive driving leading to a dramatic increase in fuel consumption and emissions, compared to normal driving. CO₂ emissions increased faster during acceleration and deceleration, and the effect of subtle changes in operating conditions on motor vehicle carbon emissions was significant. Due to traffic congestion and traffic delays caused by traffic congestion in a highway reconstruction work zone, the forced merging caused by lane closures will cause drivers to accelerate and decelerate frequently, which will greatly increase the

level of CO₂ emissions in the work zone and bring about environmental pollution. Based on a case study of a highway work zone in the United Kingdom, Huang et al. [17] conducted a complete life cycle analysis of vehicle carbon emissions resulting from traffic delays during its maintenance process. The results showed that the additional carbon emissions caused by the traffic closure due to the maintenance construction reached 400 kg and that most of the energy consumption and CO₂ emissions during the life cycle of the pavement were generated by the use of vehicles due to the nature of the pavement being used for a long time. For many years, the macroscopic model based on the average travel speed has been the most commonly used method for estimating vehicle emissions. Zeng [18] considers the internal combustion engine characteristics and traction characteristics during vehicle travel to further derive the CO₂ emission model based on vehicle travel speed and acceleration, which is more consistent with the vehicle trajectory data characteristics.

During the highway construction, the closure of some lanes in one direction will make a significant difference in the traffic flow through each lane in both directions, traffic congestion, and increased pollution emissions in the direction of the work zone. For this situation, the optimal organization can be adopted in the work area with the help of opposite lanes. The setting of reversible lanes not only avoids reducing the speed of traffic flow but also relieves traffic congestion [19]. Therefore, setting reversible lanes is an effective way to alleviate traffic congestion and vehicle pollution emissions in the work zone of highway reconstruction and expansion. Reversible lanes settings are usually studied using a bi-level programming model. Lu [20] used a bi-level programming model to minimize the total queue length of intersections and used reversible lanes to improve the space utilization of urban road intersections. In 2015, Pyakurel [21] studied the reversible lane setting problem from the evacuation perspective to evacuate people in disaster areas using the shortest time, with a successively proposed continuous time dynamic reversible lane model and its solution algorithm [22–24]. In 2019 [25], the method of reversible lane setting in urban road networks was proposed based on a microscopic simulation platform. Conceição et al. [26] considered that setting the reversible lane in a mixed traffic flow situation can substantially reduce travelers' travel time and delay in inner city applications. Arezoo et al. [27] proposed a bi-level programming model considering road closures to propose alternative routes for travelers to reduce travel time for travelers. In addition, a set of simulation experiments were conducted using the Tarrant County road network in North Texas. The results showed that the model could improve the overall road network access performance. Di [28] proposed an optimal design of reversible lanes from a system perspective based on a bi-level programming model that maximizes the coupling metric. Hong et al. [29] constructed a bi-level programming model for the use of dedicated lanes during the Winter Olympics to reduce travel costs, improve the overall road network traffic efficiency, and validate the algorithm with the help of the actual measurement data of Lian Shi Road.

In summary, previous research on road network optimization has mostly focused on the road characteristics of urban roads, such as bus lane setting and tidal lane setting, and less consideration was given to the road characteristics of highways. The objective function used in the previous bi-level programming models mostly takes the total travel time of the system as the main objective, and the CO₂ emission of the system is less embedded to the objective function.

Based on the previous research results, this paper innovatively takes the work zone of the highway, as the research object from the perspective of traffic network optimization and adds the index of total CO₂ emission of the system to the objective function. By solving the bi-level programming model embedded in the system CO₂ emissions, a reversible lane setting scheme is given for the highway work zone under three closed conditions. It provides a theoretically useful scheme for reducing pollution emissions and improving travel efficiency in the highway work zone, enriches the traffic network optimization research system, and makes an effective attempt to reduce pollution emissions in the bottleneck sections of the highway.

3. Research Methods

3.1. Structure of the Road Network

In the transport network $V = (W, A)$, W denotes all the nodes of the road network, A is all the road sections in the road network. Note that the road section $m \in A$ of the reverse section for $\bar{m}$, the number of lanes of the road section m is n_m , the single lane capacity is c_m , the road section flow is x_m . For the work zone to occupy the opposite lane driving the road section m , the number of lanes increased to u_m , $u_m = -u_{\bar{m}}$. Assume that the road traffic network V is not set up the only one-way road, the vehicle in the road section according to two-way traffic separation, u_m value should meet the relationship of $1 \leq n_m + u_m \leq n_m + n_{\bar{m}} - 1$.

3.2. The Upper-Level Model

The bi-level programming model is modeled according to a game model, where the upper-level model tends to model from the system's perspective and the lower-level model tends to model from the perspective of each individual in the system [30]. The reversible lane settings during highway reconstruction and expansion can be studied with the help of a two-level planning model. From the overall road network perspective, the objective of reversible lane setting in this model is to minimize the sum of travel time and CO₂ emissions. Travelers select travel options and minimum travel costs based on reversible lane settings. All travelers' choices affect the state of the road network, which, in turn, affects management decisions. Therefore, the goal of setting reversible lanes during highway expansion is to guide travelers to make travel choices that are beneficial to the overall state of the road network in this way.

3.2.1. System Travel Time

Let the average driving cost be t_m , calculated by using the BPR function [31]. x_m is the traffic flow of the relevant road sections. The number of lanes borrowed from the opposite direction is defined as u_m .

$$t_m(x_m, u_m) = t_{m0} \left(1 + \alpha \left(\frac{x_m}{c_m(n_m + u_m)} \right)^\beta \right) \quad (1)$$

where t_{m0} represents the road section free driving time, α , β represent the regression coefficient. Bureau of Highways model recommended coefficient $\alpha = 0.15$, $\beta = 4$ can be used. Overall travel time for Z_1 .

$$Z_1 = \sum_{m \in A} t_m(x_m, u_m) x_m \quad (2)$$

3.2.2. System Carbon Dioxide Emissions

Compared with the CO₂ emission model, which only considers the speed index, this paper adopts the CO₂ emission model based on the speed and acceleration of vehicle driving per unit time, which considers the characteristics of the internal engine and traction force during the vehicle driving process [18], and this model is consistent with the features of vehicle trajectory data and is suitable for the CO₂ emission calculation of traffic flow. Since the primary vehicles driving on the highway are small cars and large trucks, the difference in CO₂ emissions between them are large, the difference in CO₂ emissions between large trucks and small cars are considered, and the CO₂ emissions of large trucks and small cars are calculated separately.

f_m represents the individual vehicle fuel consumption rates. θ is the vehicle driving slope. This is the average acceleration of vehicles. λ is the CO₂ emission conversion coefficient for small cars-large trucks. $\beta_1 - \beta_6$ are the CO₂ emission modeling coefficients. E_{CO_2} is defined as a vehicle unit time emission. The difference in vehicle types can lead to a significant difference in their emissions. Therefore, vehicle emissions are divided into large

truck emissions and car emissions, which is defined as Z_2 and Z_3 . Moreover, the speed of trucks and cars is classified as v_a and v_b . f_{m1} and f_{m2} represent the trucks fuel rates and cars fuel rates, respectively.

$$f_m = \beta_1 \cos(\theta)v + \beta_2 \sin(\theta)v + \beta_3 v^3 + \beta_4 av + \beta_5 a + \beta_6 \quad (3)$$

$$E_{CO_2} = 2.32 \frac{f_m}{v} \quad (4)$$

$$Z_2 = 2.32 \sum_{m \in A} \frac{f_{m1}}{v_a} \cdot t_m \cdot \frac{x_m}{2} \quad (5)$$

$$Z_3 = 2.32 \lambda \sum_{m \in A} \frac{f_{m2}}{v_b} \cdot t_m \cdot \frac{x_m}{2} \quad (6)$$

3.2.3. Objective Function

The upper-level objective function of the model takes into account the total travel time of travelers and minimizes the total CO₂ emissions of large trucks and small cars in the road network objective and optimizes the number of lanes for the sections of the network that meet the conditions for setting reversible lanes in the work zone of highways.

$$\min Z = \min(Z_1 + Z_2 + Z_3) \quad (7)$$

3.3. The Lower-Level Model

The lower-level model considers the travel time of each traveler; based on the first principle of Wardrop [32], it is assumed that the traveler has a clear idea of the time spent and the choice of each route and that the traveler always chooses the shortest route. The traveler then adjusts his travel according to the road lane settings to minimize travel costs. The lower-level model allocates the roadway traffic on the lane adjustment scheme determined by the upper-level model. x_m in the upper-level planning is obtained by solving the lower-level planning, and the obtained traffic equilibrium results are used to evaluate the upper-level decision scheme. Let all travelers have the tendency to choose the path with minimum travel cost. After time adjustment, their path selection behavior will eventually satisfy the first principle of Wardrop. The road network achieves a user equilibrium state. Let the set of origin–destination pairs of OD demand be RS , for $(r, s) \in RS$, let the path set be P_{rs} , the traffic demand be q_{rs} , and the path flow of path $k \in P_{rs}$ is f_{rs}^k .

$$\min \sum_{m \in A} \int_0^{x_m} t_m(w, u_m) d_w \quad (8)$$

The variables are constrained to always keep the traffic demand for each roadway equal to the actual roadway path flow as the vehicles travel through the network, that is, the inflow–outflow equilibrium. The path flow is always positive or zero. In addition, the total traffic flow should be equal to the sum of each roadway path flow.

$$s.t. \sum_{k \in P_{rs}} f_{rs}^k = q_{rs}, (r, s) \in RS \quad (9)$$

$$f_{rs}^k \geq 0, k \in P_{rs}, (r, s) \in RS \quad (10)$$

$$x_a = \sum_{(r,s) \in RS} \sum_{k \in P_{rs}} f_{rs}^k \delta_{rs}^{ak}, a \in A \quad (11)$$

All Symbols and definitions in this part are shown in Appendix A.

4. Solving Algorithms

Since the bi-level programming model is a non-convex, non-differentiable function, the traditional mathematical solution method is complicated, so the intelligent optimization

algorithm is introduced to solve it. At present, the main intelligent algorithms currently used to solve the bi-level programming model are the genetic algorithm, simulated annealing algorithm, and particle swarm algorithm. A genetic algorithm has an excellent global search ability, which can quickly search out the whole solution in the solution space and can be easily distributed. However, the local search ability of the genetic algorithm is poor, which leads to the simple genetic algorithm being more time consuming and the search efficiency is low in the late evolutionary stage. In practical applications, genetic algorithms are prone to the problem of premature convergence. However, the simulated annealing algorithm can get rid of the optimal local solution and find out the global minimum of the objective function in the sense of probability with the stochastic search technique. However, the simulated annealing algorithm does not know much about the condition of the whole search space. It does not facilitate the search process into the most promising search region, making the simulated annealing algorithm operationally inefficient and slow in evolution. The particle swarm algorithm is fast and efficient, suitable for real-valued processing, but it is not good for discrete optimization problems and quickly falls into local optimum.

Compared with the algorithms mentioned above, the quantum particle swarm optimization algorithm converges faster and has higher intelligence and ensures the global convergence of the algorithm, so this paper designs and improves the quantum particle swarm algorithm according to the bi-level programming model.

4.1. Upper-Layer Model Solving Based on Quantum Particle Swarm Algorithm

In the traditional particle swarm optimization algorithm, the intersection of particles is mainly realized in the form of orbits. During the search process, due to the speed limit of the particles, the search area of each particle is not infinite, resulting in the algorithm in the evolution process—early convergence and trapping in local optima. To better seek the optimal global solution, Sun [33] introduced the relevant concepts of quantum mechanics into the particle swarm evolution process, and the wave function in quantum mechanics can be used to describe the state of particle motion, replacing the way of describing the position and velocity of the particle motion state in the traditional particle swarm optimization algorithm, as the particles are subject to quantum confinement, prompting them to appear at any spatial point from a probability point of view, so it is necessary to search the entire solution space, thus improving the convergence speed, population diversity, and global search capability of the algorithm [34].

$$Z_{t+1}^i = P_{i,j(t)} \pm \alpha |mbest_{j(t)} - x_{i,j(t)}| \times \ln \left[\frac{1}{u_{i,j(t)}} \right] \quad (12)$$

$$u_{i,j(t)} \sim u(0, 1) \quad (13)$$

Here, $i = 1, 2, \dots, m$, m is the particle population size. α is the shrinkage expansion factor, p_t^i denotes the historical best position corresponding to the i th particle at time t , $mbest_t$ refers to the average best position at this point, $u_t^i \sim u(0, 1)$, and g_t is the historical best position of the particle population at time t . The upper objective function of the model is used as the fitness function. According to the two-way unbalanced traffic phenomenon in the work zone of the partial lane closure and the road network conditions of the highway, m variable lanes with opposite travel lanes are randomly taken, and the quantum particle swarm algorithm solves the upper layer model in the bi-level programming model. The total travel time and total CO₂ emissions of the system are compared with the total travel time and total CO₂ emissions obtained by the optimization algorithm. If the combined total travel time and total CO₂ emission indexes are reduced, it indicates that traffic congestion and emission pollution are effectively reduced, so the optimal solution for lane setting in the non-construction direction of the borrowing lane in the work zone of the expansion can be identified. The flow chart of the algorithm is shown in Figure 1.

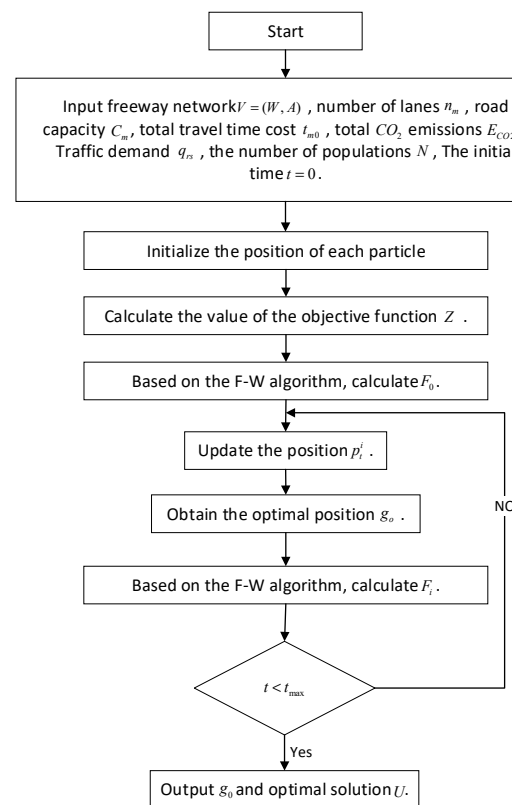


Figure 1. The quantum particle swarm algorithm flow chart.

4.2. Lower-Layer Model Solving Based on the Frank–Wolfe Algorithm

The Frank–Wolfe (FW) algorithm is the classical algorithm for solving traffic assignment problems. It allows traffic to be assigned to the path of least cost by an all-or-nothing method so that the current roadway flow can find the roadway cost. The feasible direction of descent is determined by the difference between the solution obtained by solving the subprocess and the current solution [35]. The FW algorithm repeatedly loads “all-or-nothing” on the shortest path between each origin and destination pair, converting a nonlinear planning problem into a linear one without requiring much storage capacity. Therefore, it has a significant advantage over other algorithms. The path-based traffic assignment model is obtained by bringing the applicable restrictions into the objective function.

$$\min Z(f) = \sum_{a \in A} \int_{p \in P} f_p \delta_{op} t_a(\omega) d\omega \quad (14)$$

$$s.t. \begin{cases} \sum_{p \in P_w} f_p = q_w, \forall w \in W \\ f_p \geq 0, \forall p \in P \end{cases} \quad (15)$$

The gradient of this function is as follows:

$$\frac{\partial Z(f)}{\partial f_p} = \sum_{a \in A} \delta_{op} t_a \left(\sum_{p \in P} f_p \delta_{op} \right) = \sum_{a \in A} \delta_{op} t_a(x_a) = c_p \quad (16)$$

Let $f^{(k)} = [f_p^{(k)}, p \in P]$ be a set of feasible solutions of the model, a collection of viable path flows, where $x^{(k)} = [x_a^{(k)}, a \in A]$ is a vector composed of road section flows and $c^{(k)} = [c_p^{(k)}, p \in P]$ is a vector consisting of corresponding path travel times:

$$x_a^{(k)} = \sum_{p \in P} f_p^{(k)} \delta_{ap}, \forall a \in A \quad (17)$$

$$c_p^{(k)} = \sum_{a \in A} t_a(x_a^{(k)}) \delta_{ap}, \forall a \in A \quad (18)$$

Converting the above nonlinear model to a linearly similar model:

$$\min \tilde{Z}(f) = \sum_{p \in P} c_p^{(k)} f_p = \sum_{(r,s) \in W} \sum_{p \in P^{rs}} c_p^{(k)} f_p \quad (19)$$

$$s.t. \begin{cases} \sum_{p \in P^{rs}} f_p = q^{rs}, \forall (r,s) \in W \\ f_p \geq 0, \forall p \in P \end{cases} \quad (20)$$

$$\varphi(\lambda) = Z(x^{(k)} + \lambda d_a^{(k)}) = \sum_{a \in A} \int_0^{x_a^{(k)} + \lambda d_a^{(k)}} t_a(\omega) d\omega \quad (21)$$

$$\varphi'(\lambda) = \sum_{a \in A} d_a^{(k)} t_a(x_a^{(k)} + \lambda d_a^{(k)}) = 0 \quad (22)$$

$\lambda^{(k)}$ is the root of the equation $\varphi'(\lambda) = 0$, which is the optimal step size along the feasible descent direction $d^{(k)}$. The dichotomous method is used to solve the basis of $\varphi'(\lambda)$. The above model is used to calculate the average travel cost of all vehicles in the road network, and the specific algorithm flow is shown in Figure 2.

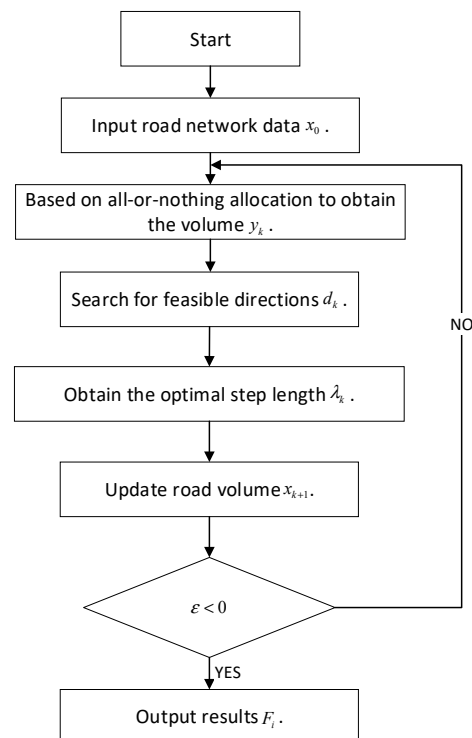


Figure 2. The Frank–Wolfe algorithm flow chart.

5. Example Simulation

5.1. Scene Building

According to the conditions related to reversible lane setting, the Sioux Falls network is used as the base road network for this paper with the help of the Sioux Falls network, which originates from Sioux Falls, the largest city in South Dakota, USA. It is one of the most commonly used networks in traffic assignment. The network includes 24 nodes and 38 two-way roads with a total of 76 road segments, of which seven components meet the requirement of borrowing opposite lanes for highway reconstruction work zones, as shown in Figure 3. The peak hour OD demand for each road section is shown in Table 1.

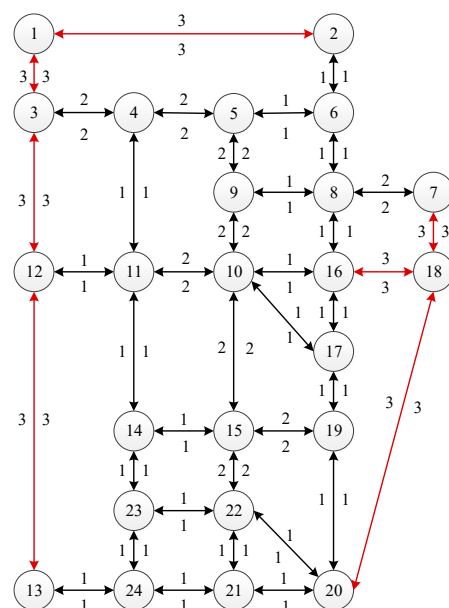


Figure 3. The Sioux Falls network structure.

Table 1. The Sioux Falls network section OD demand matrix (pcu/h).

	1	2	3	4	5	6	7	8	...	24
1	0	100	100	500	200	300	500	800	...	100
2	100	0	100	200	100	400	200	400	...	0
3	100	100	0	200	100	300	100	200	...	0
4	500	200	200	0	500	400	400	700	...	200
5	200	100	100	500	0	200	200	500	...	0
6	300	400	300	400	200	0	400	800	...	100
7	500	200	100	400	200	400	0	1000	...	100
8	800	400	200	700	500	800	1000	0	...	200
...	...	...	...	...	...	...	...	...	...	...
24	100	0	0	200	0	100	100	200	...	0

The vehicle driving speed in the highway reconstruction work zone is generally lower than that of the general road section [31,36]. Under the different construction scenarios of closing one lane, closing two lanes, and closing three lanes of the two-way three-lane road network, respectively, the average driving speed of small cars is 60 km/h. The average acceleration due to forced merging is 6 m/s², and the average driving speed of large trucks is 40 km/h, and the average acceleration due to forced merging is 4 m/s². The average driving speed of a standard section of highway without the work zone is 102 km/h for a small car and 78 km/h for a large truck, and the acceleration is 0. The total system time and total system CO₂ emissions are calculated for construction and non-construction conditions, respectively.

5.2. Two-Way 3-Lane Closure 1 Road Construction

Under the conditions of two-way three-lane closure of one-lane construction, the quantum particle swarm algorithm is used to solve the algorithm for seven two-way six-lane road sections in the road network, where the number of particles is 10 and the number of iterations is 50. the number of iterations and variations are shown in Figure 4.

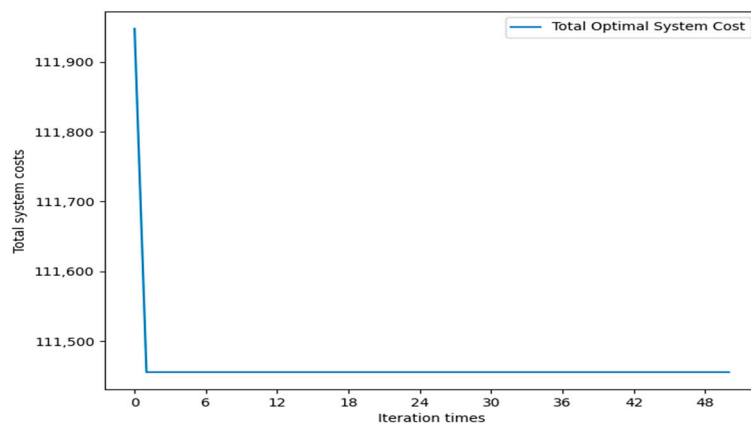


Figure 4. The algorithm iteration chart of two-way 3-lane closure 1 road construction (total system costs, s).

When each of the seven two-way three-lane roads is closed in one direction for construction without borrowing the opposite lane, the total travel cost is 113,022.9. The total travel time cost is 11,047.2, the total CO₂ emission cost of small cars is 42,896.3, and the total CO₂ emission cost of large trucks is 59,079.4.

After the optimization model solution, the lane setting scheme for borrowing the opposite lane is obtained, which is shown in Figure 3 and Table 2. The total travel cost after borrowing the opposite lane is 111,455.7, of which the total travel time cost is 10,885.8, the total CO₂ emission cost of small cars is 42,313.1, and the total CO₂ emission costs of large trucks is 58,256.9. The optimized road network of two-way 3-lane closure 1 road construction and reversible lane setting scheme of two-way 3-lane closure 1 road construction are shown in Figure 5 and Table 2.

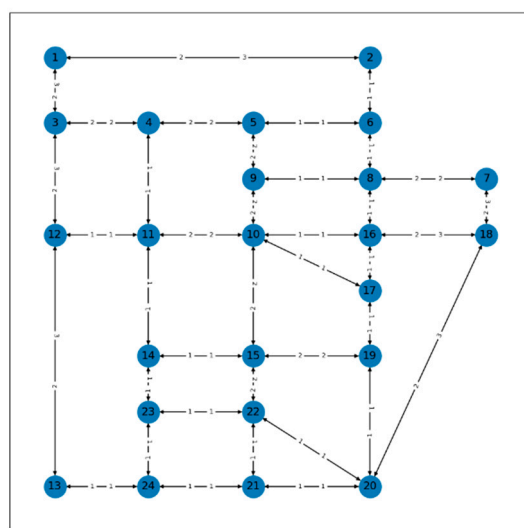


Figure 5. The optimized road network of two-way 3-lane closure 1 road construction (optimized numbers of lanes in each direction).

Table 2. Reversible lane setting scheme of two-way 3-lane closure 1 road construction.

Road Sections	1-2	1-3	3-12	7-18	12-13	16-18	18-20
Number of lane closures	1	1	1	1	1	1	1
Number of reversible lanes	0	1	1	1	1	0	1

Two-way three-lane closure of one road construction, borrowing the opposite lane and not borrowing the opposite lane travel cost comparison is shown in Table 3.

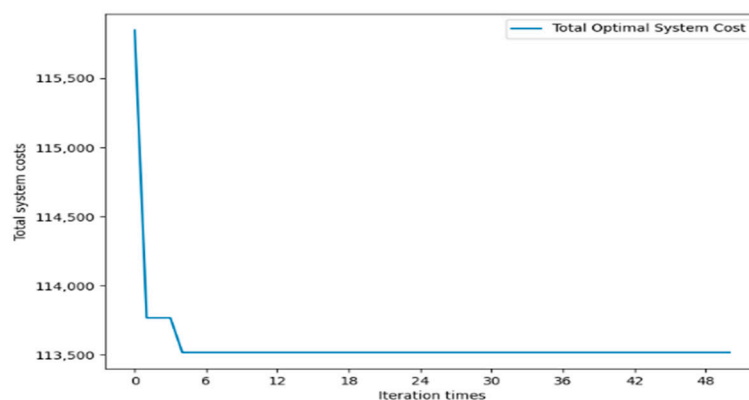
Table 3. Comparison of optimized solution and original condition (s).

Closure of 1 Lane	Z/s	Z ₁ /s	Z ₂ /s	Z ₃ /s
Original condition	113,022.9	11,047.2	42,896.3	59,079.4
Optimized solution	111,455.7	10,885.8	42,313.1	58,256.9
Change rate	−1.39%	−1.46%	−1.36%	−1.39%

As seen from the above table, the total system cost and time and emissions of setting up the opposite lane have been reduced. However, due to the slight reduction, the number of lanes borrowed from the work zone is small and the construction cost generated by the possible borrowing of opposite lanes is considered. Considering the actual construction difficulty, it is likely not to provide a reversible lane if only one lane is closed during construction.

5.3. Two-Way 3-Lane Closure 2 Road Construction

The quantum particle swarm algorithm is used to solve the algorithm for seven two-way six-lane road sections in the road network under the conditions of two-way three-lane closure and two-lane construction, where the number of particles is 10 and the number of iterations is 50. The number of iterations and variations are shown in Figure 6.

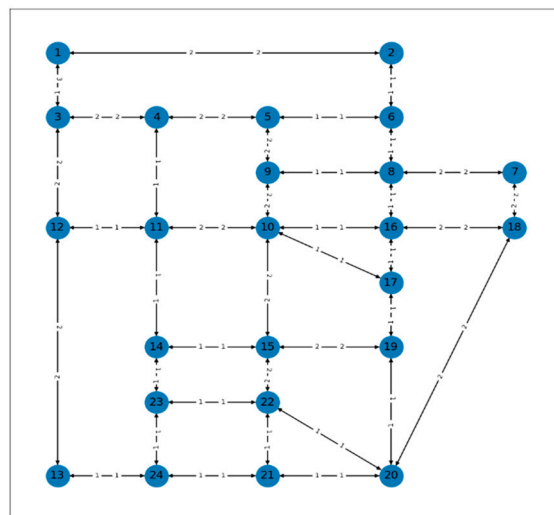
**Figure 6.** The algorithm iteration chart of two-way 3-lane closure 2 road construction (total system costs, s).

The total travel cost is 126,072.7 when each of the seven two-way three-lane road sections is closed in one direction for two roads for construction without borrowing the opposite lane. A total of 12,365.4 of the total travel time cost, 47,789.6 of the total CO₂ emission cost of small cars, and 65,917.8 of the total CO₂ emission cost of large trucks.

After the optimization model is solved, the lane setting scheme for driving in the opposite lane is obtained, as shown in Figure 5 and Table 4. The total travel cost after borrowing the opposite lane is 113,518.7, of which the total travel time cost is 11,094.9, the total CO₂ emission cost of small cars is 43,085.5, and the total CO₂ emission cost of large trucks is 59,338.2. The optimized road network of two-way 3-lane closure 2 road construction and reversible lane setting scheme of two-way 3-lane closure 2 road construction are shown in Figure 7 and Table 4.

Table 4. Reversible lane setting scheme of two-way 3-lane closure 1 road construction.

Road Sections	1-2	1-3	3-12	7-18	12-13	16-18	18-20
Number of lane closures	2	2	2	2	2	2	2
Number of reversible lanes	1	2	1	1	1	1	1

**Figure 7.** The optimized road network of two-way 3-lane closure 2 road construction (optimized numbers of lanes in each direction).

Two-way three-lane closed 2 road construction, borrowing the opposite lane and not borrowed opposite lane travel cost comparison is shown in Table 5.

Table 5. Comparison of optimized solution and original condition (s).

Closure of 2 Lane	Z/s	Z ₁ /s	Z ₂ /s	Z ₃ /s
Original condition	126,072.7	12,365.4	47,789.6	65,917.8
Optimized solution	113,518.7	11,094.9	43,085.5	59,338.2
Change rate	−9.96%	−10.27%	−9.84%	−9.98%

From the above table, it can be seen that the total system cost, time cost and emissions are significantly reduced after setting reversible lanes, and the reduction is about 10%. Therefore, in the case of closing two lanes, the corresponding number of reversible lanes should be set in sections 1-2, 1-3, 3-12, 7-18, 12-13, 16-18 and 18-20.

5.4. Two-Way 3-Lane Fully Closed Construction

Under the condition of two-way three-lane fully closed construction, the quantum particle swarm algorithm is used to solve the algorithm for seven two-way six-lane road sections in the road network, where the number of particles is 10 and the number of iterations is 50. The number of iterations and variations of the upper model in the two-layer planning model are shown in Figure 8.

When each of the seven two-way three-lane road sections is fully enclosed in one direction and uses the opposite lane for driving, the optimization model is solved to obtain the lane setting scheme for driving by the opposite lane, as shown in Figure 7 and Table 6. The total travel cost after borrowing the opposite lane is 120,862.8, of which the total travel time cost is 11,810.5, the total CO₂ emission cost of small cars is 45,876.0, and the CO₂ emission cost of large trucks is 63,176.2. The optimized road network of two-way three-lane fully closed construction and comparison of optimized solution and original condition are shown in Figure 9 and Table 6.

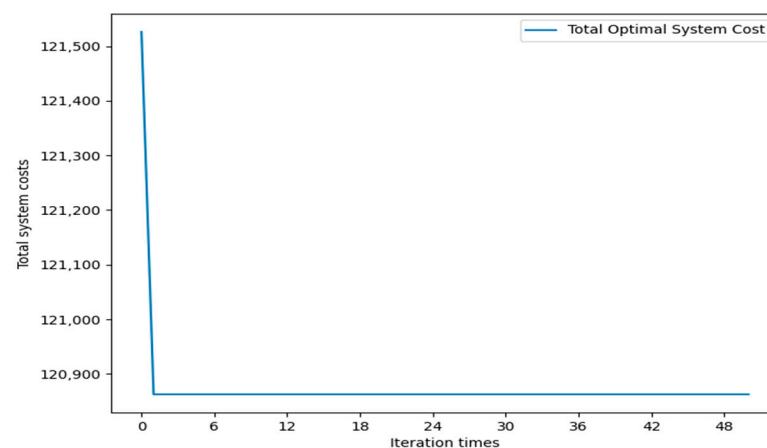


Figure 8. The algorithm iteration chart of two-way three-lane fully closed construction (total system costs, s).

Table 6. Comparison of optimized solution and original condition.

Road Sections	1-2	1-3	3-12	7-18	12-13	16-18	18-20
Number of lane closures	3	3	3	3	3	3	3
Number of reversible lanes	2	2	2	2	1	2	2

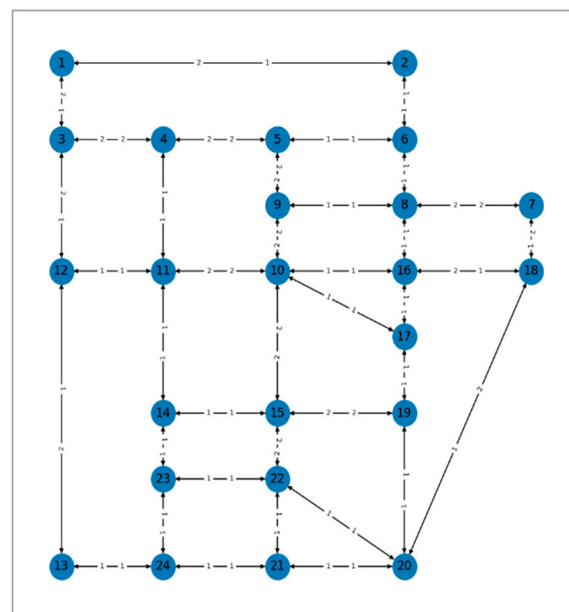


Figure 9. The optimized road network of two-way three-lane fully closed construction (optimized numbers of lanes in each direction).

When a measured road is fully closed, the opposite lane must be borrowed to ensure the normal movement of vehicles. Therefore, in the case of three-lane closure, using the two-layer planning model, the solution is derived from its optimal reversible lane setting scheme as shown in the table above, which should be set in 1-2, 1-3, 3-12, 7-18, 12-13, 16-18, 18-20 sections of the corresponding number of reversible lanes.

5.5. Comparison of Optimized Solutions under Different Closure Conditions

The optimization model was solved for different closure conditions of the work zone of the two-way six-lane highway reconstruction and expansion. The optimized

costs under three closure conditions are shown in Table 7. The total cost, time cost, and CO₂ emission cost are the smallest among the three closure scenarios, which is the most economical construction plan. The total cost, time cost, and CO₂ emissions cost after the model optimization of the closed two lanes construction, compared with not borrowing the opposite lane, decreased the most, so the completed two lanes construction program should actively choose to borrow the opposite lane optimization method, and the construction period is short, fully closed construction conditions; the number of borrowed opposite lanes can refer to the above optimization program values, according to the actual road factors, vehicle factors, and economic cost factors for appropriate setting. Therefore, according to the location, function, duration, and solid construction type of the construction zone, and to minimize the traffic congestion and environmental problems caused by the work zone, we can choose to set reasonable reversible lanes according to the above study.

Table 7. Comparison of optimized costs under three closure conditions (s).

Three Closure Conditions	Z/s	Z ₁ /s	Z ₂ /s	Z ₃ /s
Closure of 2 lane	111,455.7	10,885.8	42,313.1	58,256.9
Closure of 2 lane	113,518.7	11,094.9	43,085.5	59,338.2
Fully closed	120,862.8	11,810.5	45,876.0	63,176.2

6. Conclusions

This paper addresses the traffic congestion in the highway work zone and the environmental pollution caused by the congestion and it proposes that the measures related to the opening of opposite lanes can still be used when the road is not fully closed. A new approach and theoretical basis are proposed to solve the problem of traffic congestion in temporary bottleneck areas of highways. This paper promotes the integration of knowledge in the environmental and transportation fields and makes a valuable exploration of the sustainable development of transportation and the environment. At the same time, we propose a policy related to the general standard two-way six-lane highway work zone. In the two-way six-lane highway for construction, priority should be given to vehicle travel time. When the number of closed lanes is greater than one, the opposite lane should be opened according to the traffic volume, which can significantly reduce the total system travel cost and the total system CO₂ emissions and minimize the work zone on the overall traffic operation status and environmental impact.

Regarding the technical means, we take the system vehicle travel time as the basis, consider the different CO₂ emission characteristics of large trucks and small cars, and construct CO₂ emission models for large trucks and small vehicles, respectively. The objective function is innovatively constructed by minimizing travel time costs and CO₂ emissions. Compared with the objective function considering only the total travel time, the influencing factor of CO₂ emissions on the system travel cost are added, which is a helpful exploration to reduce the system's CO₂ emissions. The quantum particle swarm algorithm is used to solve the objective function, which uses the wave function in quantum mechanics to describe the state of particle motion, replacing the position and velocity description of particle motion state in the basic particle swarm optimization algorithm, so as to avoid falling into local optimum to the maximum extent, and has better global search capability and higher search efficiency, compared with a genetic algorithm, particle swarm algorithm, and simulated annealing algorithm. The results of our research show that different traffic volumes, large-truck-to-small-bus ratios, and the number of lane closures in the traffic network will affect the CO₂ emission costs and total system time costs. Therefore, the applicability of our results to each country and regions highway work zones requires further consideration of the traffic volumes, the ratio of large trucks to small buses, and the number of construction lane closures in the traffic networks of different countries and regions. First, future work can be conducted from the mathematical modeling point of view to add factors, such as construction period and construction cost for highway work zones, and further consider the construction of objective functions under different

traffic networks and a different large-truck-to-small-bus ratios and the traffic safety factors and the economic costs required to open opposite lanes, as well as the intervention of autonomous vehicles, to further improve the lane adjustment scheme of highways work zones. The work zone restrictions are further improved according to the relevant policies and regulations of different countries and regions. Second, subsequent research avenues can also use econometric models for safety risk analysis and in-depth assessment of the risk factors associated with road construction in work zones [37]. Safety indicators can be introduced into the bi-level programming model, as well as CO₂ emission indicators to further improve the mathematical model. Third, other road traffic bottlenecks similar to highway work zones can also be extended and explored with the technical means of this paper. Finally, the limitation of our research is the simulated environment instead of an actual environment. If better experimental conditions are available, some actual experiments can be carried out in the future.

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Appendix A

Table A1. Symbol and definition.

Symbol	Definition
v	the transport network
W	all nodes of the transport network
A	the sections of the transport network
$\bar{m}$	the reverse sections of the transport network
n_m	the number of lanes on the side of the work zone
c_m	the single-lane capacity
x_m	the traffic flow of the relevant road sections
u_m	the number of lanes borrowed from the opposite direction
t_m	the average travel time
t_{m0}	the free travel time
α	regression coefficients of the BPR function
β	regression coefficients of the BPR function
θ	the vehicle driving slope
γ	the average speed of vehicles
a	the average acceleration of vehicles
λ	the CO ₂ emission conversion factor for small cars-large trucks
$\beta_1 - \beta_6$	the CO ₂ emission modeling factors
RS	the set of origin–destination pairs of OD demands
P_{RS}	the path set
q_{rs}	the traffic demand
f_{rs}^k	the path flow of k

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